3. Problem Description

This study investigates a multi-robot task allocation and routing problem arising in smart farming environments, where task locations are represented as rectangular regions rather than point targets. Each region requires a robot to perform a full coverage operation using a back-and-forth sweeping motion, subject to restricted robot kinematics and geometric constraints. All robots are assumed to be homogeneous and operate in the same planar workspace. They move at a constant speed v. When switching the sweeping direction between consecutive passes, a fixed turning time τ is incurred.

Unlike classical multi-robot routing problems that assume fixed service times at task nodes, this study explicitly models the internal motion of robots within task regions, where processing time and exit location are endogenously determined by region geometry, sweep width, and robot motion patterns.

3.1 Task Regions and Spatial Representation

The system consists of a set of task regions located in a two-dimensional plane. For any task i, its region is defined by the coordinates of two diagonal vertices: the lower-left corner $(x_i^L,\ y_i^B)$ and the upper-right corner $(x_i^R,\ y_i^T)$, where $x_i^L < x_i^R$, and $y_i^B < y_i^T$. Therefore, each task region has four vertices, namely the left-bottom (LB), left-top (LT), right-bottom (RB), and right-top (RT) corners. The robot is allowed to enter and leave a task region only through these four vertices. Entering or exiting from any other location on the boundary or within the interior of a region is not permitted.

3.2 Restricted Robot Movements and Coverage-Based Processing within Task Regions

Upon entering a task region, a robot must perform a full coverage operation using a sweeping motion. A fixed sweep width δ is given, representing the width covered by the robot in each sweeping pass. The robot's movements and operations inside a task region are subject to the following restrictions:

1. Axis-aligned robot motion
   Within a task region, the robot is restricted to moving horizontally or vertically only.
2. Sweep mode selection
   For each task region, the robot must select either a Horizontal Sweep Mode (H-Mode) or a Vertical Sweep Mode (V-Mode):
   2.1. Horizontal Sweep Mode (H-Mode): The robot performs back-and-forth horizontal sweeps parallel to the x-axis. The number of sweeping passes is computed as:

$$N_i^H = \left\lceil \frac{y_i^T - y_i^B}{\delta} \right\rceil$$

   The total in-region travel distance under H-Mode is

$$D_i^H = N_i^H \times (x_i^R - x_i^L)$$

   2.2. Vertical Sweep Mode (V-Mode): The robot performs back-and-forth vertical sweeps parallel to the y-axis. The required number of sweeping passes in this mode is:

$$N_i^V = \left\lceil \frac{x_i^R - x_i^L}{\delta} \right\rceil$$

   The total in-region travel distance under V-Mode is

$$D_i^V = N_i^V \times \left(y_i^T - y_i^B\right)$$

3. Computation of in-region processing time

The processing time is endogenously determined by the region dimensions, sweep width, robot speed, and turning time, which is defined as follows,

$$T_i = \begin{cases} \frac{D_i^H}{v} + (N_i^H - 1) \times \tau, & if\ H\text{-}Mode\ is\ selected \\ \frac{D_i^V}{v} + (N_i^V - 1) \times \tau, & if\ \text{V-}Mode\ is\ selected \end{cases}$$

4. Coupling between entry vertex, sweep parity, and exit vertex

The exit vertex of a task region is no longer a free decision variable, but is implicitly determined by the selected entry vertex, the sweep mode, and the parity of the number of sweeping passes.

4.1. In H-Mode:

a) If the robot enters from a bottom-side vertex (LB or RB), it exits from a top-side vertex (LT or RT).

b) If the robot enters from a top-side vertex (LT or RT), it exits from a bottom-side vertex (LB or RB).

Therefore, if N_i^H is odd (even), the exit vertex lies on the opposite (same) horizontal side relative to the entry vertex.

4.2. In V-Mode:

a) If the robot enters from a left-side vertex (LB or LT), it exits from a right-side vertex (RB or RT).

b) If the robot enters from a right-side vertex (RB or RT), it exits from a left-side vertex (LB or LT).

Thus, if N_i^V is odd (even), the exit vertex lies on the opposite (same) vertical side relative to the entry vertex.

3.3 Grid-based Robot Movements between Task Regions

After completing the processing operation in a task region and leaving it through the determined exit vertex, the robot moves to the next task region. For movements between different task regions, the robot is restricted to axis-aligned motion. Therefore, the travel distance between two task regions is measured as the Manhattan distance between the exit vertex of the current region and the entry vertex of the next region.

3.4 Task Execution and Multi-Robot Routing Decisions

Each task region must be visited and fully covered by exactly one robot. For each assigned task region, the robot must jointly determine:

a) The entry vertex,

b) The sweep mode (horizontal or vertical),

c) The number of sweeping passes implied by the region geometry and sweep width,

d) The resulting exit vertex determined by sweep parity.

The visiting sequence of task regions for each robot and the inter-region travel paths must also be determined. All robots are assumed to start from a common depot and return to the same depot after completing their assigned task regions. The objective is to minimize the makespan, defined as the maximum completion time among all robots, where the completion time of each robot is

measured from its departure from the depot to its return to the depot after finishing all assigned coverage tasks and inter-region movements.

3.5 Problem Characteristics

Compared with conventional robot routing problems, the proposed problem exhibits the following additional characteristics:

a) Processing time inside a task region is endogenously determined by geometric dimensions, sweep width, robot speed, and turning time, rather than being a fixed parameter.

b) The exit vertex is implicitly determined by entry location and sweep parity, which further strengthens the coupling between routing and processing decisions.

c) The introduction of sweep width and coverage-based motion leads to a tighter integration between continuous geometric properties and discrete routing decisions.

These characteristics significantly increase the structural complexity of the problem and motivate the development of advanced modeling and solution approaches.

The depot (initial location) is located at the coordinate (0, 0).

Robot moving speed $v = 1$.

A fixed sweep width $\delta = 2$.

A fixed turning time $\tau = 0.5$.

A simple example:

Task	(x_L, y_B)	(x_R, y_T)	w (width)	h (height)	N^H	T^H	N^V	T^V
1	(8, 6)	(16, 9)	8	3	2	16.5	4	13.5
2	(20, 4)	(24, 16)	4	12	6	26.5	2	24.5
3	(30, 10)	(42, 14)	12	4	2	24.5	6	26.5
4	(12, 20)	(18, 34)	6	14	7	45.0	3	43
5	(26, 22)	(34, 30)	8	8	4	33.5	4	33.5
6	(40, 22)	(46, 38)	6	16	8	51.5	3	49

Task 1 / 2 / 3 / 5

a) H-mode: using the even-parity mapping
LB→LT, RB→RT, LT→LB, RT→RB

b) V-mode: using the even-parity mapping
LB→RB, LT→RT, RB→LB, RT→LT

Task 4

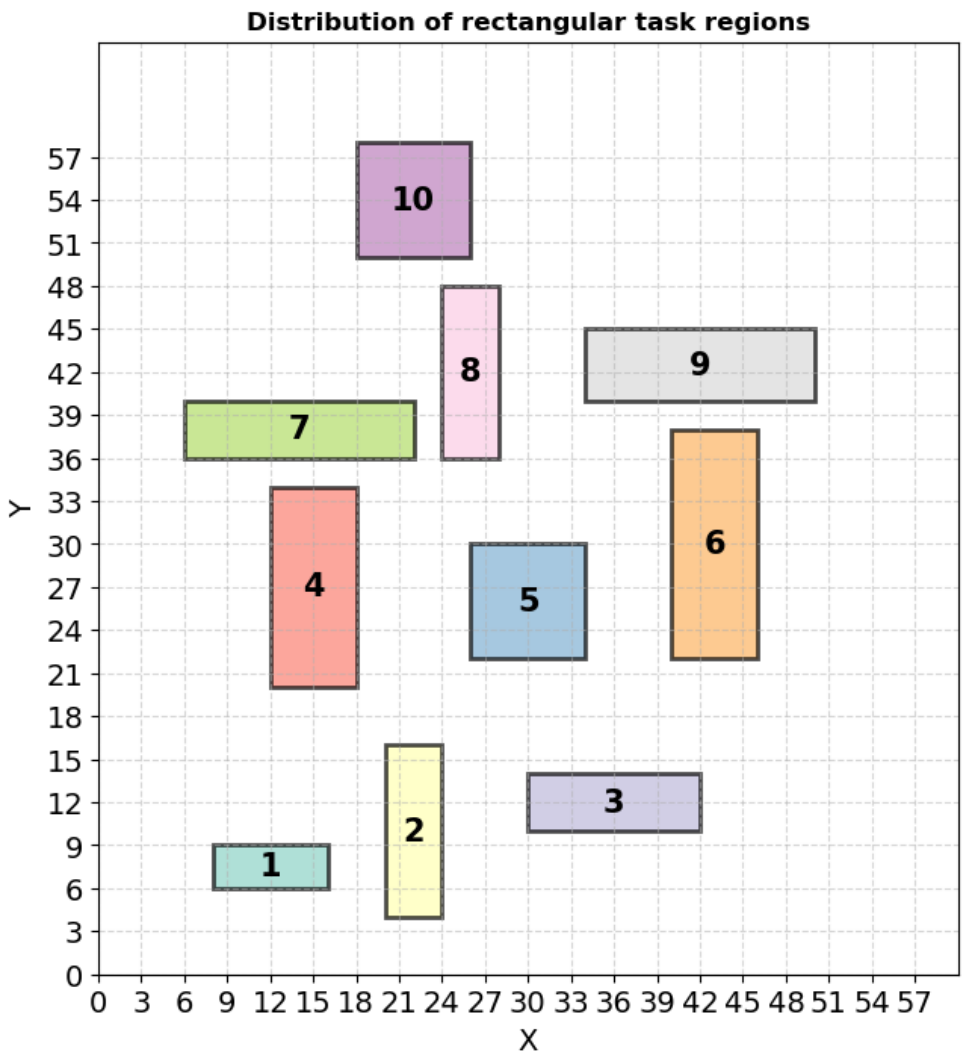
a) H-mode: using the odd-parity mapping
LB→RT, RB→LT, LT→RB, RT→LB

b) V-mode: using the odd-parity mapping
LB→RT, LT→RB, RB→LT, RT→LB

Task 6

a) H-mode: using the even-parity mapping
LB→LT, RB→RT, LT→LB, RT→RB

b) V-mode: using the odd-parity mapping
LB→RT, LT→RB, RB→LT, RT→LB


Distribution of rectangular task regions
10
8
9
7
6
4
5
3
2
1
Y
X
0 3 6 9 12 15 18 21 24 27 30 33 36 39 42 45 48 51 54 57
0 3 6 9 12 15 18 21 24 27 30 33 36 39 42 45 48 51 54 57